Set 30: Mobile App Usage Analysis

AIM

To analyze mobile application usage using R programming by creating a histogram and density plot of screen time, a scatter plot of data usage versus screen time, and a bar chart comparing average satisfaction by gender.

ALGORITHM

1. Create a dataset containing User ID, Gender, Age, Screen Time, App Usage Count, Data Used, Satisfaction, and Usage Date.
2. Create a histogram and density plot for Screen Time.
3. Create a scatter plot between Data Used and Screen Time.
4. Calculate the correlation coefficient and add a regression line.
5. Calculate average Satisfaction for each gender.
6. Create a bar chart displaying average satisfaction with value labels.
7. Interpret mobile usage and satisfaction patterns.

R PROGRAM

This program also uses base R only.

```
app <- data.frame(
  ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Count = c(18,25,12,30,10,22),
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction = c(3,5,3,5,2,4),
  Usage_Date = as.Date(c("2025-01-08","2025-01-08",
                         "2025-02-11","2025-02-11",
                         "2025-03-14","2025-03-14"))
)


hist(app$Screen_Time, probability = TRUE,
     col = "skyblue", main = "Screen Time Distribution",
```

```
    xlab = "Screen Time (Hours)")
lines(density(app$Screen_Time), col = "red", lwd = 2)

plot(app$Screen_Time, app$Data_Used,
    pch = 19, col = "blue",
    main = "Screen Time vs Data Usage",
    xlab = "Screen Time (Hours)", ylab = "Data Used (GB)")
abline(lm(Data_Used ~ Screen_Time, data = app),
       col = "red", lwd = 2)

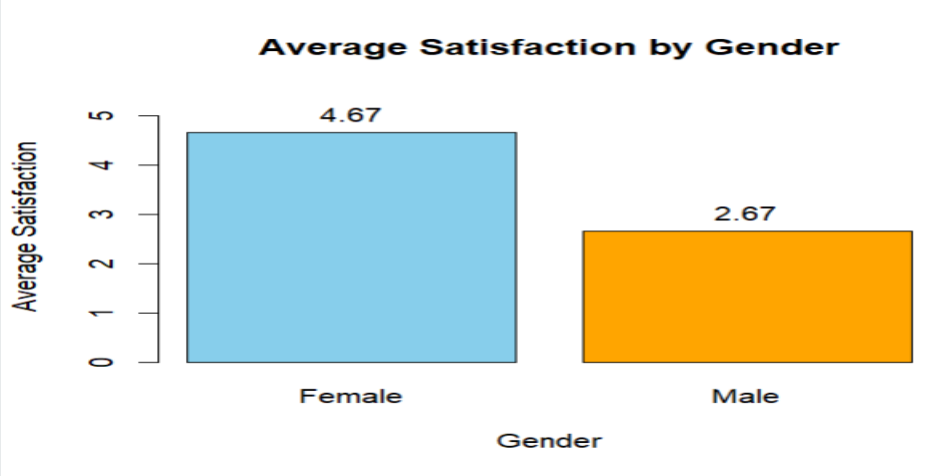
print(cor(app$Screen_Time, app$Data_Used))

avg <- tapply(app$Satisfaction, app$Gender, mean)

bp <- barplot(avg, col = c("skyblue", "orange"),
              main = "Average Satisfaction by Gender",
              xlab = "Gender", ylab = "Average Satisfaction",
              ylim = c(0, 5.5))
text(bp, avg, labels = round(avg, 2), pos = 3)

print(avg)
```

OUTPUT:



RESULT

The histogram and density plot show the distribution of mobile app screen time. The scatter plot and regression line examine the relationship between screen time and data usage. The correlation coefficient measures the strength and direction of this relationship.

Average satisfaction scores:

Gender	Average satisfaction
Female	4.67
Male	2.67

Female users have a higher average satisfaction score in this sample. The scatter plot is expected to show a positive relationship between screen time and data usage.